IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

OPTIMAL CONTROL

Thursday, 16 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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OPTIMAL CONTROL

1. Find the equation of the curve $y(x)$ that is an extremal for the functional

$$J(y) = \int_0^{x_f} \frac{\sqrt{1+(y'(x))^2}}{y(x)}\,dx \qquad (x_f > 0)$$

for the boundary conditions $y(0) = 0$ and the point $(x_f, y(x_f))$ varying along the line $y = x - 5$.

a) Write the Lagrangian of the problem. [3]

b) Write the first-order necessary condition for optimality given by the Euler-Lagrange equation. [6]

c) Compute the general form of the extremal curve. [4]

d) Calculate the extremal curve by using the boundary conditions. [7]

2. A scalar linear system is described by the differential equation

$$\dot{x}(t) = u(t) \qquad\qquad x(0) = 0$$

It is desired to find the control law that minimises the performance measure

$$J(u) = x(t_f)^2 + \int_0^{t_f} \left((x(t) - \cos t)^2 + u(t)^2 \right) dt$$

The admissible state and control values are not constrained, the final time t_f is specified, and $x(t_f)$ is free.

a) Construct the Hamiltonian of the problem. [3]

b) Write the necessary conditions for optimality given by the maximum principle. [5]

c) Write the boundary conditions that must be satisfied. [4]

d) Calculate the optimal costate trajectory as a function of time t and constant parameters. [4]

e) Compute the optimal state trajectory and the optimal control law as a function of time t. [4]

3. Consider the second order linear system with one control variable described by the following equations

$$\dot{x}_1(t) = x_2(t) + u(t)$$
$$\dot{x}_2(t) = -u(t)$$

which is to be controlled so that the following cost functional is minimised

$$J(u) = \frac{1}{2}\int_0^{t_f} x_1(t)^2 dt$$

with $x_1(0)$, $x_2(0)$ and t_f given, $x_1(t_f) = x_2(t_f) = 0$, and $u(t) \in [-M, M]$ for all t, where M is a positive constant.

a) Construct the Hamiltonian of the problem. [2]

b) Compute the switching function. [2]

c) Calculate the optimal control in terms of the switching function. [2]

d) Derive the adjoint equations. [2]

e) Show that singular arcs exist and compute the singular control law. [7]

f) Calculate the locus in the state space where singular arcs exist. [5]

4. A double integrator is described by the following equations

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = u(t)$$

It is desired to drive the double integrator to the origin $(x_1(t_f) = x_2(t_f) = 0)$ by a control law that minimises the cost

$$J(u) = \int_0^{t_f} \left(x_1(t)^2 + x_2(t)^2 + u(t)^2\right) dt$$

The terminal time t_f is free and the admissible control values are not constrained.

a) Write the associated Hamilton-Jacobi-Bellman (HJB) differential equation. [3]

b) Compute the control in terms of the value function that achieves the infimum in the HJB equation and use the computed control to simplify the HJB equation. [4]

c) Write the boundary condition that must be satisfied by the value function. [3]

d) Assume that the HJB differential equation admits a solution of the form

$$V(t,x) = \sqrt{3}x_1(t)^2 + ax_1(t)x_2(t) + bx_2(t)^2$$

where a and b are unknown constants that are to be found.

i) Calculate a and b so that the assumed solution satisfies the HJB differential equation and the boundary condition found in part c). [7]

ii) Compute the optimal control as a state feedback law. [3]